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HarvestPay



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1 INTRODUCTION

1.1 Description of the problem area

The HarvestPay project is designed for the agricultural sector, specifically for viticulture (grape growing), which is a very important industry in the Maribor region. Grape harvesting is a short but very intense period that requires many temporary workers to be managed at once.

1.1.1 Chronological overview and precursors

Historically, tracking the harvest and the work done relied on manual, paper-based methods. In the first stage, managers at weighing stations used physical clipboards and pens to write down how many buckets each picker brought. This old way was very slow, and it was easy to make mistakes or lose the papers in the field. In the second stage, many vineyards started using Excel spreadsheets. While this was a step forward, it still required hours of manual typing at the end of the day to move data from paper to the computer, creating big administrative delays and slowing down payments to workers.

1.1.2 Why the product is better and problems it solves

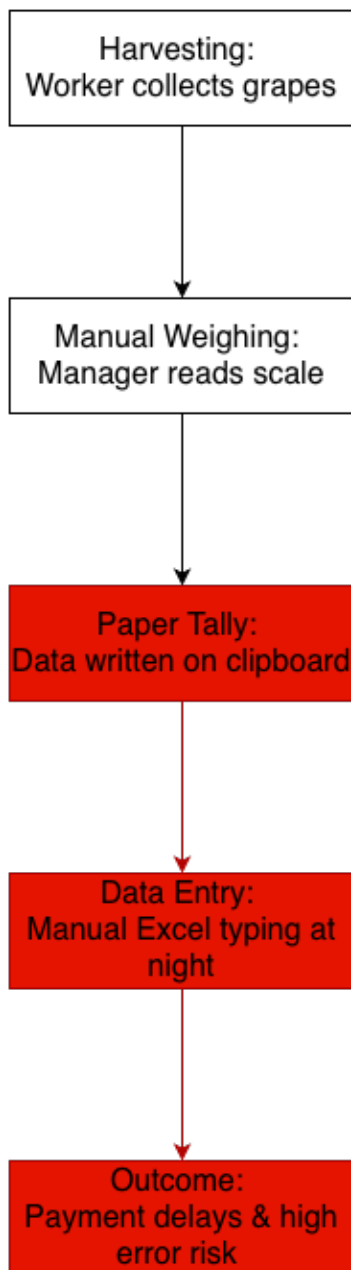
Using a modern, IoT-based system like HarvestPay with NFC wristbands and smartphone scanning is a much better approach because it solves several practical and social problems:

- **No more manual typing:** Scanning an NFC wristband logs the data instantly into the cloud. This saves time, removes human errors, and stops the need for paper records.
- **Real-time information:** Vineyard managers can see exactly how many tons of grapes have been picked at any moment of the day, helping them plan the work better.
- **Protection for workers:** In the EU, many seasonal workers are at risk of being underpaid or working "off the books". By using a digital system, every bucket picked

is recorded permanently. This ensures transparent and fair pay for everyone and helps prevent labor exploitation

The transition from the old manual process to our digital solution is shown in the flowchart in Figure 1.1.

Past: Manual & Paper-based precursors



Present/Future: HarvestPay IoT Solution

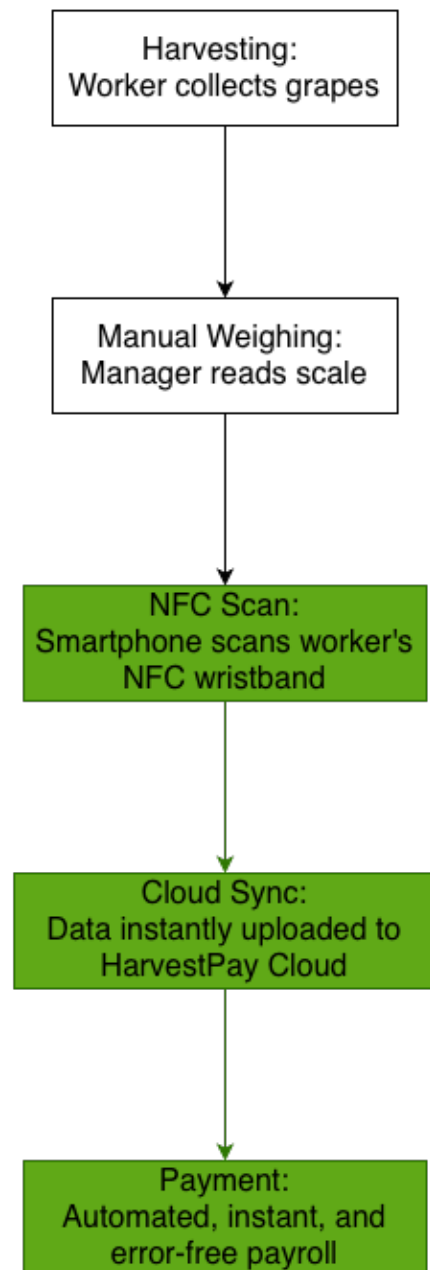


Figure 1.1: Presentation of application areas - Chronological flowchart of harvest logistics

1.2 Brief description of the idea

The method is based on digital identification and harvest recording. Each worker is assigned an NFC wristband linked to their profile in the system. At the weighing point, the supervisor scans the wristband, records the harvested quantity, and the app automatically stores the data and calculates the worker's pay according to the predefined rate. The most common use cases include worker registration, NFC-based worker identification, harvest quantity recording, automatic payment calculation, monitoring of harvest progress, overview of worker performance, and export of payroll data.

Example:

A vineyard hires 20 seasonal workers during the harvest season and pays them based on the quantity of grapes they harvest. Each worker receives an NFC wristband with a unique identifier. After a worker brings the harvested grapes for weighing, the supervisor scans the NFC wristband, confirms the measured weight in the app, and the system automatically assigns the quantity to that specific worker. For example, if the scanned worker harvested 120 kg of grapes in one day and the payment rate is €0.35 per kilogram, HarvestPay immediately calculates that the worker earned €42. All records are stored digitally, so the manager can track total harvest output, individual worker performance, and total payroll in real time.

1.3 Literature review

Recent studies emphasize the urgent need for digital transformation in agriculture to optimize workflows and internal logistics. Traditional management methods in viticulture have historically relied on manual data collection, which significantly reduces overall efficiency [1]. As noted by Remondino and Zanin [2], the digitization of agri-food logistics is currently a key factor in increasing competitive advantage and business sustainability. Automating routine operations and implementing tracking technologies allow companies to accelerate supply chains and substantially lower operational costs [2].

Furthermore, the implementation of digital tracking systems has critical social and legal significance. According to an analytical briefing by the European Parliament [3], the EU

agricultural sector is highly dependent on the labor of seasonal workers, and the level of undeclared (informal) employment in the industry can exceed 60%. Traditional paper and cash-based payment methods often lead to the exploitation of vulnerable workers and the rise of illegal intermediaries (such as the *caporalato* system), where a portion of the earnings is illegally withheld. The introduction of transparent solutions like HarvestPay fully aligns with modern Agriculture 4.0 standards and the EU's calls for the protection of workers' rights. The use of digital monitoring technologies, such as NFC wristbands, not only makes operational management more transparent but also guarantees fair compensation, eliminating human error and financial abuse by intermediaries [1], [3].

1.4 Limitations

HarvestPay is limited to harvest tracking and related payment calculation, not for full vineyard management. While our main focus is on vineyards, the app can easily be used for any other fruit harvest, like apples or berries. The smartphone requires an internet connection (4G/5G or Wi-Fi) to synchronize the collected data with the cloud. While the app is designed to work in the field using an offline mode, the final data sync and office updates depend on network availability.

The solution does not include:

- Recruitment
- Contract administration
- GPS tracking of workers
- Warehouse stock management
- Advanced and AI analytics

2 PROJECT BRIEF

2.1 Project Definition

The main goal of this project is to build a digital system called HarvestPay that makes grape harvesting easier to manage. We want to replace the old way of writing things on paper with a modern solution using NFC (Near Field Communication) technology. The project needs to provide a mobile app for managers to scan workers' wristbands and a cloud database to store all the harvest data. The final goal is to have a system that calculates exactly how much each worker should be paid based on the weight of grapes they collect, making the whole process fast and transparent.

2.2 Background

We came up with the idea for HarvestPay while looking at how local wineries here in the Maribor region handle their harvests. Even though modern wineries use very expensive machines for making wine, the way they manage seasonal workers is still very "old school." We saw managers struggling with paper clipboards in the rain and wind, and we heard stories about mistakes in counting buckets that lead to arguments with workers. We realized that almost every manager has a smartphone, and NFC tags (like in credit cards) are very cheap. We thought: why not combine them? We wanted to bring "Agriculture 4.0" to the local vineyards in a simple way that actually works for real people.

2.3 Project objectives

To make this project a success, we have set the following performance goals:

- **Time:** The software must be fully developed and tested within 4 months, so it is ready before the next harvest season begins.
- **Cost:** We want to keep the development budget under €60,000. The system should be affordable for medium-sized family vineyards.

- **Quality:** The mobile app must be "field-proof." It needs to be very simple so that a manager can learn to use it in just a few minutes, and the buttons must be large enough to use while working outside.
- **Scope:** We will deliver a mobile scanning app (Android/iOS) and a web-based dashboard for the winery office.
- **Risk:** The biggest risk is the lack of internet in some parts of the vineyard. We will include an "offline mode" so the manager's smartphone can save data locally and sync it with the cloud as soon as a connection is restored.
- **Benefit:** The system should reduce the time spent on evening paperwork by at least 70% and completely remove human errors in payroll.

2.4 Desired outcomes

By the end of this project, we expect to achieve the following results:

- **The HarvestPay Software Suite:** A fully functional mobile application for scanning and a secure web-based dashboard for managers to monitor everything in real-time.
- **Administrative Efficiency:** We want to eliminate the "evening rush" where managers spend hours typing data from paper to Excel. The data will already be in the system, ready to use.
- **Transparency and Trust:** Using a digital system means no more "he said, she said" arguments. If a worker picks 120 kg, they can see exactly that amount (and the calculated €42) on the screen immediately. This creates a much fairer work environment for seasonal staff.
- **Better Data for Wineries:** The owners will get clear stats on which vineyard rows are the most productive and which workers are the fastest. This helps them plan future harvests much better than they could with just paper notes.

2.5 Project scope and exclusions

The scope of this project covers the development of the Android/iOS scanning app, the setup of the secure cloud database, and a web dashboard for the winery office to monitor operations and export payroll data in .csv or .pdf formats. We will also provide a short

training manual for the staff. As part of the project, we include hardware sourcing and validation: we have researched and found a preferred supplier for durable, waterproof NFC wristbands (NTAG213 chips). We will provide the winery with the exact specifications and a verified purchasing link to ensure the hardware is 100% compatible with our software. However, to keep the project realistic, we have some clear exclusions. We are not covering the actual purchase costs of the hardware; the winery is expected to buy the required smartphones and NFC wristbands themselves using our recommended supplier. Also, while the system calculates the exact payroll, it won't actually send money directly to bank accounts - that is still a job for the winery's accountant using our generated reports. Lastly, we decided not to include real-time GPS tracking of workers in order to save smartphone battery life during long shifts and to avoid complicated privacy and GDPR issues.

2.6 Constraints and assumptions

The farm has an NFC-compatible mobile device for scanning NFC wristbands. The device must have periodic internet connection to sync data. Each worker will be assigned a functioning NFC wristband linked to their profile. Each worker has a profile with current and correct information already created in the system before starting work.

2.7 Project tolerances

A schedule tolerance of one to two weeks is acceptable if unexpected delays occur during development or testing, with the final version of the system still delivered before the harvest starts. If the system is not able to receive data directly from the weighing device, manual entry of the measured weight by the supervisor is considered acceptable. We allow a 10% budget overspend (approx. €1,200). This buffer is for unexpected costs, such as higher prices for durable NFC wristbands or extra cloud hosting fees. In terms of quality tolerance, the system must have 99% accuracy in scanning and data logging.

2.8 The user(s) and any other known interested parties

The main users of the HarvestPay system are vineyard managers, supervisors, and seasonal workers. Supervisors will use the mobile application at the weighing point to scan NFC wristbands and record harvested quantities. Managers and winery owners will use the web

dashboard to monitor worker performance, harvest progress, and payroll data. Seasonal workers are also important users, because the system directly affects how their work is recorded and how their payment is calculated. Other interested parties include the winery accountant, who will use the exported payroll data for final wage processing, and the technical support responsible for maintaining the system.

2.9 Outline Business Case

The HarvestPay system is needed because many vineyards still rely on paper-based methods for recording harvest quantities and calculating worker payment. This traditional process is slow and creates confusion or disagreements between workers and management as workers have limited visibility into how their pay is calculated. This solution was chosen because it is practical, affordable, and easy to use in real vineyard conditions. Compared with paper records, it offers faster data entry, more accurate worker identification, immediate wage calculation, and better access to harvest data. The expected business benefits include reduced administrative workload, fewer payroll mistakes, improved transparency, and better harvest planning through accurate data.

2.10 Project Product Description

The main project product is the HarvestPay digital system, which consists of a mobile application for NFC scanning and harvest entry, a cloud database for storing records, and a web-based dashboard for monitoring and exporting data. The system will allow supervisors to identify workers by scanning their NFC wristbands, record the quantity of harvested grapes, and automatically calculate payment based on a predefined rate. Managers will be able to review individual and total harvest records, monitor worker productivity and export payroll-related data.

The customer's quality expectations are that the system should be reliable and easy to use. The mobile app should have a simple interface, fast response time, and clear data display so that supervisors can use it with minimal training.

The user acceptance criteria are that the app must successfully scan NFC wristbands and identify the correct worker in at least 99% of test cases. It must record harvested quantities and calculate payment with 99% accuracy according to the predefined payment rate. The dashboard must display all stored records correctly and allow export of payroll data in at least one required format, such as CSV or PDF. The operations and maintenance acceptance criteria are that the system must run on NFC-compatible smartphones with Android or iOS support and must synchronize data with the cloud successfully whenever an internet connection is available. Authorized users must be able to update worker profiles, change payment rates, and correct wrongly entered records through the system.

3 ORGANIZATIONAL STRUCTURE

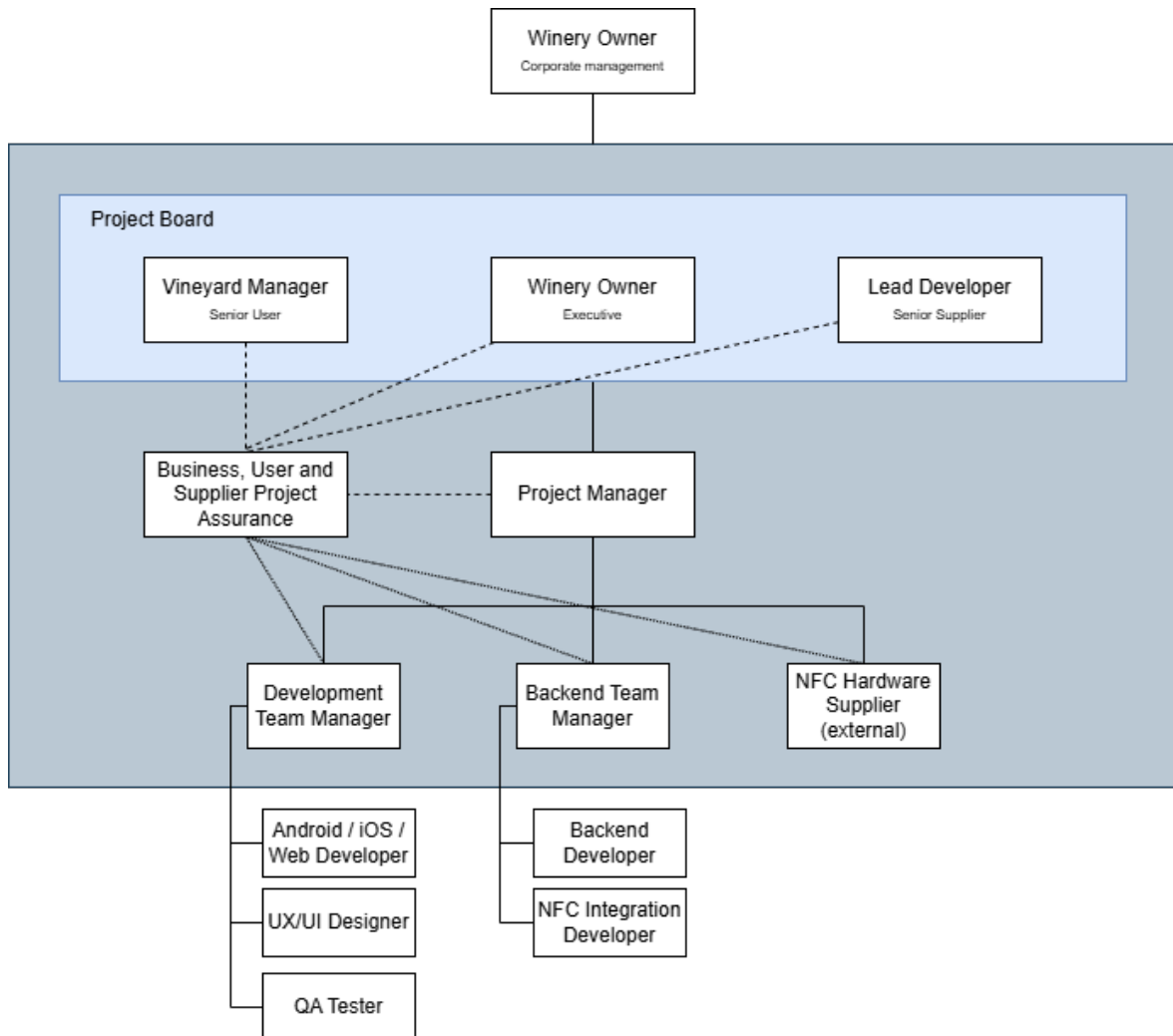


Figure 3.1 HarvestPay Project organizational structure

Role	Responsibility
Winery Owner (Corporate management)	Provides the project mandate.
	Approves the project at the highest level.
	Ensures that the project supports the strategic and operational needs of the winery.
Executive (Winery Owner)	Represents the business interest, owns the business case.
	Approves key project decisions.

Senior User (Vineyard Manager)	Represents the needs of end users.
	Ensures that the system supports vineyard operations.
	Defines user requirements and confirms that the delivered solution is usable in practice.
Senior Supplier (Lead Developer)	Represents the supplier side.
	Provides technical resources and expertise, and confirms that the product can be delivered at the required quality level.
Business, User and Supplier Project Assurance	Monitors whether the project remains viable from business, user and supplier perspectives.
	Checks that requirements are met and provides independent assurance to the Project Board.
Project Manager	Manages the project on a day-to-day basis.
	Plans and coordinates activities, controls scope, time, cost, quality, risks and communication and reports progress to the Project Board.
Development Team Manager	Manages the development team responsible for the application interface and front-end functionality.
	Assigns tasks, monitors progress and ensures that outputs meet agreed requirements and quality standards.
Backend Team Manager	Manages backend and system integration work.
	Coordinates database and API-related tasks, ensures correct system logic and oversees technical integration between app, backend and NFC functionality.
NFC Hardware Supplier (external)	Provides compatible NFC hardware (wristbands).
	Ensures that wristbands meet technical requirements.
	Supports hardware-related consultation or adjustments if needed.
Android / iOS / Web Developer	Develops the application interface and functionality for mobile and web environments.
	Implements required features and supports issue fixing after testing.
UX/UI Designer	Designs the user interface and user experience.
	Ensures that the application is simple and practical for field conditions and supports usability improvements.

QA Tester	Tests the system, identifies defects, verifies whether requirements are fulfilled.
	Supports quality assurance before deployment.
Backend Developer	Develops the server-side logic, database communication, APIs and core processing functions such as harvest record storage and payroll calculation.
NFC Integration Developer	Implements and tests NFC scanning functionality.
	Ensures correct communication between the app and NFC wristbands.
	Supports integration with backend services.

Table 3.1. Responsibilities relevant to the Project Organizational structure

3.1 Justification for Excluded Roles

The Change Authority: Given the medium project budget (€60,000) and the very short development timeframe (4 months), a separate Change Authority has been excluded to avoid unnecessary bureaucracy and administrative delays. Instead, the decision-making power for all changes remains with the Project Board to maintain agility and ensure rapid responses to any technical or operational issues that may arise during the intense harvest-preparation cycle without the need for additional committees.

Project Support: The project scale and limited budget do not justify the cost of a dedicated administrative resource. Consequently, all essential administrative tasks -including maintaining the issue log, file management, and scheduling - are delegated to and handled directly by the Project Manager. This ensures that overhead costs remain low while maintaining accurate project documentation and coordination.

4 RISK REGISTER / RISK LOG

Identified risk	Risk author	Risk owner	Date	Risk description	Risk category	Probability	Proximity	Risk response actions	Risk status
Lack of internet connectivity in the vineyard	Project team	Project Manager		Vineyards are located in rural areas with limited mobile network coverage. Weak or unstable internet connection may prevent real-time synchronization of harvest data. Data is only stored locally, risking loss if the device is damaged or lost before syncing.	Technical / operational / infrastructure	High	Near	1. Implement offline mode with local storage on device. 2. Synchronize automatically when connection is restored. 3. Test app in low-signal conditions.	
NFC wristband scanning failure	Lead Developer	NFC Hardware Supplier		NFC wristbands may not be scanned correctly due to hardware incompatibility, damaged tags, or reading errors, causing delays in worker identification.	Technical / operational / infrastructure	High	Near	1. Test multiple wristband types. 2. Provide winery with verified supplier.	
Incorrect worker profile data	Project team	Vineyard Manager		If worker profiles contain incorrect or incomplete information, harvested quantities may be assigned to the wrong person, leading to payroll errors.	Organizational / management / human factors	Medium	Near	Verify profiles before harvest starts; assign clear responsibility for data entry; allow corrections by authorized users; review records regularly.	
Manual entry error of harvested weight	Project team	Vineyard Manager		If the weight cannot be transferred automatically and must be entered manually, the supervisor may record a wrong quantity, affecting payment calculation.	Technical / operational / infrastructure	Medium	Near	1. Use clear input screen. 2. Confirm entries before saving 3. Allow authorized correction of records. 4. Provide short training for supervisors.	
Delay in software development	Project team	Project Manager		Development tasks may take longer than planned due to integration problems or limited resources, which may delay testing and final delivery before the harvest season.	Organizational / management / human factors	Medium	Medium	1. Use phased planning. 2. Monitor milestones and prioritize core features. 3. Apply schedule tolerance of one to two weeks if necessary.	

Identified risk	Risk author	Risk owner	Date	Risk description	Risk category	Probability	Proximity	Risk response actions	Risk status
Budget overspend	Project team	Executive		The total project cost may exceed the planned budget due to additional development time, higher cloud costs or more expensive NFC hardware.	Financial	Low	Medium	1. Monthly budget review against actuals. 2. 10% pre-approved budget tolerance (report to Project Board if tolerance is close to be exceeded). 3. Use cost-effective tools and services where possible.	
Low user acceptance	Project team	Vineyard Manager		Managers or supervisors may resist using the new system if they find it complicated or less practical than paper-based methods.	Organizational / management / human factors	Medium	Near	1. Keep interface simple (UI with large buttons and minimal steps). 2. Provide on-site training session and manual before harvest begins. 3. Gather feedback before deployment.	
Hardware supplier delivery issue	Project team	Senior Supplier		The NFC wristband supplier may fail to deliver compatible hardware on time or provide lower-quality products than expected.	Strategic / commercial	Medium	Medium	Select verified supplier; order samples early; define technical requirements clearly; identify backup supplier.	
Cloud service outage or data loss	Project team	Backend Team Manager		A third-party cloud provider may experience downtime or data loss, which could make harvest records temporarily or permanently unavailable and prevent payroll from being processed, causing payment delays and possible disputes with workers.	Technical / operational / infrastructure	Very Low	Near	1. Choose a cloud provider with 99.9%+ SLA. 2. Implement daily automated database backups. 3. Use automatic sync retry.	
Data protection / GDPR issue	Project Manager	Backend Team Manager		The system stores personal data of workers in a cloud database. Data may be stored or handled incorrectly, leading to legal penalties under EU GDPR, reputational damage, and potential shutdown of the system by authorities.	Legal and regulatory	Low	Medium	1. Limit stored data to required information. 2. Obtain explicit written consent from all workers before enrollment. 3. Use secure storage. 4. Conduct GDPR compliance review before launch.	
Incorrect payroll calculation	Project team	Backend Team Manager		Errors in the calculation logic may result in incorrect payment amounts, reducing trust in the system.	Technical / operational / infrastructure	Low	Near	1. Require 99% calculation accuracy as acceptance criterion. 2. Review formulas with user representatives. 3. Provide manual correction capability for authorised users.	

Identified risk	Risk author	Risk owner	Date	Risk description	Risk category	Probability	Proximity	Risk response actions	Risk status
Key team member unavailable	Project team	Project Manager		A key project member may become unavailable due to illness, resignation, reassignment, or other circumstances, which could slow down development, testing, or coordination and negatively affect the project schedule.	Organizational / management / human factors	Medium	Medium	1. Share knowledge among team members. 2. Document work regularly. 3. Define backup responsibilities for critical roles. 4. Reassign tasks quickly if unavailability occurs.	

Table 4.1 Project Risk Register

Detailed description of selected risks:

1. Poor internet connectivity in the vineyard

This is one of the most important risks because the project depends on digital data collection and cloud synchronization. Vineyards may have weak signal coverage, especially in more remote areas. If no action is taken, supervisors may be unable to upload harvest records in real time, which could reduce the usefulness of the system during daily operations. The chosen response is to introduce offline mode, where the application temporarily stores records on the device and synchronizes them later when the connection is restored. This reduces the operational impact of connectivity problems.

2. NFC wristband scanning failure

HarvestPay relies on NFC wristbands for fast and accurate worker identification. If the wristbands are damaged, incompatible with the scanning device, or difficult to read in field conditions, the system may not identify workers correctly. This could cause delays at the

weighing point and reduce trust in the solution. To manage this risk, the project should verify compatibility before deployment, use tested wristband models, keep spare wristbands available and allow manual worker selection in exceptional cases.

3. Manual entry error of harvested weight

Even if worker identification is automated, the weight of harvested grapes may still need to be entered manually if the scale is not directly integrated with the system. In that case, the supervisor may accidentally enter the wrong value, which would affect worker payment and total harvest data. This risk is important because it directly influences the accuracy and fairness of the system. The proposed response is to simplify the input process, require confirmation before saving and enable correction of records by authorized users.

4. Low user acceptance

The technical solution can work correctly, but the project may still underperform if vineyard managers or supervisors do not want to use it. This may happen if they are used to paper-based work or if they consider the app too complicated for field conditions. If this risk materializes, the system may be ignored or only partially used. To reduce this risk, the app should have a simple interface, users should receive short training and feedback from real users should be included during testing and early deployment.

		Consequence				
		1 - Negligible	2 - Minor	3 - Moderate	4 - Major	5 - Catastrophic
Likelihood	5 - Almost certain					
	4 - Likely			[R2]	[R1]	
	3 - Possible			[R3]; [R4]; [R8]; [R12]	[R5]; [R7]	
	2 - Unlikely			[R6]	[R11]	[R10]
	1 - Rare					[R9]

Figure 4.1 Risk Assessment Matrix

[R1] Lack of internet connectivity in the vineyard	16
[R2] NFC wristband scanning failure	12
[R3] Incorrect worker profile data	9
[R4] Manual entry error of harvested weight	9
[R5] Delay in software development	12
[R6] Budget overspend	6
[R7] Low user acceptance	12
[R8] Hardware supplier delivery issue	9
[R9] Cloud service outage or data loss	5
[R10] Data protection / GDPR issue	10
[R11] Incorrect payroll calculation	8
[R12] Key team member unavailable	9

Table 4.2 Risk Scoring Summary and Quantitative Assessment

Low	
Moderate	
High	
Extreme	

Table 4.3 Risk Level
Legend

5 PROJECT PLAN

The project plan serves as a practical roadmap for the implementation of the HarvestPay system. It organizes the project into several consecutive phases, from initiation and planning through analysis, design, development, testing, deployment, support and final closure. Within each phase, the plan defines the main activities that need to be completed, their expected duration, logical dependencies and the human resources involved in their execution.

An important function of the project plan is to support control and coordination throughout the project lifecycle. By assigning start and finish dates to activities and identifying review points, the plan makes it possible to monitor whether the project is progressing according to expectations. It also helps the team recognize where activities can be performed in parallel and where one task depends on the completion of another. As a result, the project plan contributes to better time management, more efficient use of available resources and earlier identification of possible delays.

Name	Duration	Start	Finish
HarvestPay	149 days	1. 4. 2026 8:00	30. 10. 2026 17:00
Initiation and planning phase	14 days	1. 4. 2026 8:00	21. 4. 2026 17:00
Project kick-off with customer	1 day	1. 4. 2026 8:00	1. 4. 2026 17:00
Project Brief	1 day	2. 4. 2026 8:00	2. 4. 2026 17:00
Define project roles and responsibilities	1 day	3. 4. 2026 8:00	3. 4. 2026 17:00
Prepare project plan	1 day	7. 4. 2026 8:00	7. 4. 2026 17:00
Analysis of current processes	3 days	8. 4. 2026 8:00	10. 4. 2026 17:00
Definition of functional requirements	3 days	13. 4. 2026 8:00	15. 4. 2026 17:00
Definition of non-functional requirements	3 days	13. 4. 2026 8:00	15. 4. 2026 17:00
Selection of NFC supplier	2 days	13. 4. 2026 8:00	14. 4. 2026 17:00
Define communication strategy	1 day	15. 4. 2026 8:00	15. 4. 2026 17:00
Definition of acceptance criteria	0,5 days	16. 4. 2026 8:00	16. 4. 2026 13:00
Prepare risk register	0,5 day	16. 4. 2026 13:00	16. 4. 2026 17:00

Requirements validation with customer	0,5 days	17. 4. 2026 8:00	17. 4. 2026 13:00
Set up project environment - software, licenses	2,5 days	17. 4. 2026 8:00	21. 4. 2026 17:00
Analysis and design phase	16 days	17. 4. 2026 13:00	13. 5. 2026 13:00
System architecture design	6 days	17. 4. 2026 13:00	30. 4. 2026 13:00
UX/UI wireframes	6 days	17. 4. 2026 13:00	28. 4. 2026 13:00
Design review meeting 1	0,5 days	4. 5. 2026 8:00	4. 5. 2026 13:00
Cloud and database design	3,5 days	28. 4. 2026 13:00	7. 5. 2026 17:00
NFC integration design	3 days	5. 5. 2026 8:00	12. 5. 2026 17:00
Preparation of test scenarios	4 days	7. 5. 2026 13:00	13. 5. 2026 13:00
Design review meeting 2	0,5 days	13. 5. 2026 8:00	13. 5. 2026 13:00
Development phase	21 days	11. 5. 2026 8:00	8. 6. 2026 17:00
Set up development environment	3 days	11. 5. 2026 8:00	13. 5. 2026 17:00
Backend development	10 days	14. 5. 2026 8:00	27. 5. 2026 17:00
Backend development review 1	0,5 days	20. 5. 2026 8:00	20. 5. 2026 13:00
Backend development review 2	0,5 days	27. 5. 2026 8:00	27. 5. 2026 13:00
Mobile App development	10 days	14. 5. 2026 8:00	27. 5. 2026 17:00
App development review 1	0,5 days	20. 5. 2026 13:00	20. 5. 2026 17:00
App development review 2	0,5 days	27. 5. 2026 13:00	27. 5. 2026 17:00
Web development	7 days	28. 5. 2026 8:00	5. 6. 2026 17:00
Web development review	0,5 days	1. 6. 2026 8:00	1. 6. 2026 13:00
NFC integration development	7 days	28. 5. 2026 8:00	5. 6. 2026 17:00
Development review	1 day	8. 6. 2026 8:00	8. 6. 2026 17:00
Testing phase	17 days	9. 6. 2026 8:00	2. 7. 2026 17:00
NFC wristband compatibility testing	3 days	9. 6. 2026 8:00	11. 6. 2026 17:00
Mobile App and Web testing	3 days	12. 6. 2026 8:00	16. 6. 2026 17:00
Payroll calculation testing	1 day	17. 6. 2026 8:00	17. 6. 2026 17:00
User experience testing	2 days	17. 6. 2026 8:00	18. 6. 2026 17:00
Bug fixing phase 1	5 days	19. 6. 2026 8:00	26. 6. 2026 17:00
Offline mode terrain testing	1 day	29. 6. 2026 8:00	29. 6. 2026 17:00
Bug fixing phase 2	3 days	30. 6. 2026 8:00	2. 7. 2026 17:00

Deployment and training phase	14 days	30. 6. 2026 8:00	17. 7. 2026 17:00
Prepare user and training manual	3 days	30. 6. 2026 8:00	2. 7. 2026 17:00
Pilot installation and system configuration	2 days	3. 7. 2026 8:00	6. 7. 2026 17:00
User training	2 days	7. 7. 2026 8:00	8. 7. 2026 17:00
Final user feedbacks review and adjustments	3 days	9. 7. 2026 8:00	13. 7. 2026 17:00
Documentation and handover	4 days	10. 7. 2026 8:00	15. 7. 2026 17:00
Monitoring and support phase	54 days	17. 8. 2026 8:00	29. 10. 2026 17:00
Live support	20 days	17. 8. 2026 8:00	11. 9. 2026 17:00
Monitoring of system	54 days	17. 8. 2026 8:00	29. 10. 2026 17:00
Post-deployment review meeting	0,5 days	19. 8. 2026 8:00	19. 8. 2026 13:00
Critical issues fixing	2 days	19. 8. 2026 13:00	21. 8. 2026 13:00
Risk management	21,5 days	27. 5. 2026 8:00	26. 6. 2026 13:00
Milestone progress and budget review 1	0,5 days	27. 5. 2026 8:00	27. 5. 2026 13:00
Milestone progress and budget review 2	0,5 days	26. 6. 2026 8:00	26. 6. 2026 13:00
Project closure	1 day	30. 10. 2026 8:00	30. 10. 2026 17:00
Lessons learned	0,5 days	30. 10. 2026 8:00	30. 10. 2026 13:00
Project closure and final handover	0,5 days	30. 10. 2026 13:00	30. 10. 2026 17:00

Table 5.1 Project Plan

5.1 Project Plan Scenario

During the project execution, an unexpected delay was simulated in which the activity Set up development environment was extended from the originally planned 3 days to 6 days. This delay affected the start of the following development activities and had an impact on the critical path. After updating the project progress, it was necessary to evaluate whether the delay should be accepted or managed through corrective action.

The first option (Figure 5.1) is to leave the project unchanged and absorb the delay. Since the project plan contains some time reserve at the end, it is possible to lose 3 days without affecting the final delivery date. New delivery date would be postponed from 17.7.2026 to

22.6.2026. In this case, no additional staffing costs are generated and the project remains within the original budget. This is the most efficient option if the delay does not affect other critical milestones.

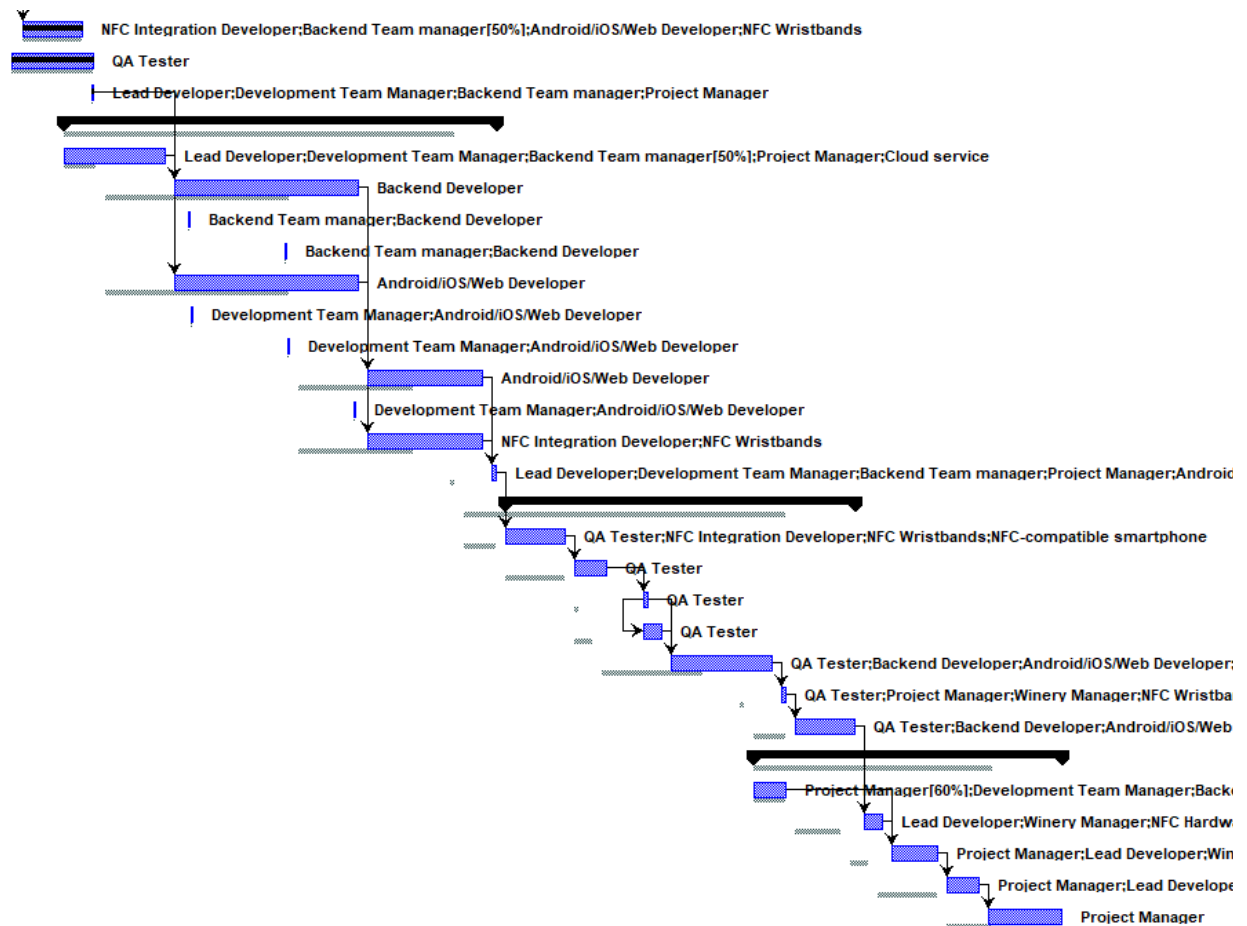


Figure 5.1 Project plan scenario with delay

The second option (Figure 5.2) is to recover the delay by adding new resources, specifically a Backend Developer and a Web Developer, in order to accelerate the following early development activities. With limited use of these additional resources, it is possible to reduce the duration of both main development activities to the required 7 days and recover the lost time. This solution makes it possible to bring the project closer to the original schedule while still remaining within the total budget limit of €60,000 (Figure 5.3).



Figure 5.2 Project plan scenario with additional resources



Figure 5.3 Project Cost with additional resources

6 CONCLUSION

The main goal of the HarvestPay project was to design a digital solution that would improve the management of seasonal grape harvesting by replacing paper-based records with a more accurate, transparent and efficient system based on NFC wristbands, mobile scanning and cloud data storage. The project was focused on supporting vineyard managers and supervisors in recording harvested quantities, calculating worker payments and monitoring harvest progress in real time. From the project management perspective, the objective was not only to propose the product itself, but also to prepare a structured and realistic project plan for its implementation within defined time, cost, quality and scope constraints.

During the project, we analysed the problem area in detail, including the traditional paper-based and spreadsheet-supported methods currently used in vineyards and we explained why these approaches are inefficient. We defined the project brief in detail, including project objectives, desired outcomes, scope and exclusions, constraints and assumptions, tolerances, business case, stakeholders and product description. This means that the project was not limited to a general idea only, but was elaborated into a realistic concept with clearly specified expectations and acceptance criteria.

We also prepared the organizational structure of the project according to PRINCE2 principles and assigned responsibilities to all important roles. Furthermore, we identified and assessed the main project risks, prepared a risk register, described selected risks in more detail and evaluated them by probability, proximity and response actions. Finally, we developed a detailed project plan covering the full project lifecycle from initiation and planning through analysis, design, development, testing, deployment, support and closure, including durations, logical dependencies and review points. Overall, the project was developed to a level of detail that makes it possible to understand not only what HarvestPay should do, but also how such a system could be realistically organized and implemented in practice.

LITERATURE AND RESOURCES

- [1] S. V. Tapkire and D. V. Nandre, "IoT-enabled automated monitoring systems in viticulture: Current trends, implementation challenges, and opportunities for improving grape crop yield," *Open Access Int. J. Sci. Eng.*, vol. 8, no. 3, pp. 53–62, 2025. [Online].
Available: https://www.oaijse.com/VolumeArticles/FullTextPDF/1019_IOT-ENABLED_AUTOMATED_MONITORING_SYSTEMS.pdf. [Accessed: 28-Mar-2026].

- [2] M. Remondino and A. Zanin, "Logistics and Agri-Food: Digitization to Increase Competitive Advantage and Sustainability. Literature Review and the Case of Italy," *Sustainability*, vol. 14, no. 2, p. 787, Jan. 2022. [Online].
Available: <https://www.mdpi.com/2071-1050/14/2/787>. [Accessed: 29-Mar-2026].

- [3] M.-L. Augère-Granier, "Migrant seasonal workers in the European agricultural sector," European Parliamentary Research Service (EPRS), Feb. 2021. [Online].
Available: [https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/689347/EPRS_BRI\(2021\)689347_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/689347/EPRS_BRI(2021)689347_EN.pdf). [Accessed: 29-Mar-2026].